



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 01



CORN POSTHARVEST PROCESSING PLANT (Alamada Site)

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Test Report No.	:	2013 - 01
Date of Test	:	January 9-11, 2013
Test Valid Until	:	January 11, 2018
Brand	:	-
Model	:	-
Type	:	Postharvest processing plant for corn
Designed & Manufactured By	:	MINDANAO MULTI- MACHINERIES & ENGINEERING WORKS and Engr. WILFREDO P. CRUZ Sayre Highway, Bangcud Malaybalay City, Bukidnon
Test Requested by:	:	Manufacturer

SUMMARY

The Plant is a postharvest processing plant for corn with an input capacity of 100 tons of ear corn per day. It was designed to dry newly harvested ear corn to moisture content of around 20%_{w.b.} using a dehusked corn dryer after which these will be shelled in corn sheller and then dried in grain dryers to a final moisture content of 14%_{w.b.}. The plant was tested for performance using 99836 kg of ear corn and using corn cobs as fuel for the heating system.

Dehusked Corn Dryer

Using the two 50-ton capacity drying bins, the Ear Corn Dryer had a Drying Capacity of 2812 kg/h of ear corn from an initial grain Moisture Content of 27.7%_{w.b.}

to a grain Moisture Content of 19.1%_{w.b.} with an average Moisture Reduction per hour of 0.24%. The actual Drying Time for the ear corn with an initial weight of 99836 kg was 35.5 hours. During the test, the Drying Air Temperature was 66.8°C. The Fuel Consumption of the ear corn dryer's biomass boiler was 197.8 kg/h of corn cob while the Power Consumption of the dryer during operation was 19.0 kW with Load Current of 39.3 A and Line Voltage of 472 V.

A Heating System Efficiency of 78.5% was obtained for the dryer. Based on the moisture content of samples taken at the end of drying, Moisture Content Variation was $\pm 0.46\%$.

Corn Sheller

The Corn Sheller was tested using the ear corn dried from the dehusked corn dryer with 19.1% MC_{wb} and Kernel-Ear Corn Ratio of 0.81.

The Corn Sheller had a Shelling Capacity of 16113 kg/h with shelled grain Purity of 99.70%. At a Shelling Cylinder Speed of 836 rpm and Peripheral Speed of 1338 m/min, Shelling Recovery was 99.73% and Shelling Efficiency was 99.91%. The machine had a Total Shelling Loss of 0.27% consisting of 0.16 % Scattering Loss, no Blower Loss, 0.02% Separation Loss and 0.09 Unshelled Loss.

For the purpose of standardization, the average Corrected Shelling Capacity was 16225 kg/h based on 20% MC_{wb} and 100% Purity.

Grain quality analysis of shelled grains showed a Net Percent Cracked Grains of 4.67 % and 3.45 % Broken Grains.

Grain Dryers

Using a single biomass boiler (corn cob furnace) to provide heat to the two 16-ton capacity grain dryers, the Grain Dryers had a Drying Capacity of 9069 kg/h from an initial grain Moisture Content of 19.1%_{w.b.} to a grain Moisture Content of 13.2%_{w.b.} with an average Moisture Reduction per hour of 1.68%. The actual Drying Time for the corn kernel with an initial weight of 31743 kg was 3.5 hours. The average Drying Air Temperature during the test was 67.9°C. The Fuel Consumption of the grain dryer's biomass boiler was 187.2 kg/h of corn cob

while the Power Consumption of the dryer during operation was 24.6 kW with Load Current of 52.2 A and Line Voltage of 461 V.

The Heating System Efficiency of the dryer was 86.8%. Heat Utilization was 5684 kJ per kg moisture removed with a corresponding Drying Efficiency of 41.2%. The dryer had a Drying System Efficiency of 35.7%. Based on the moisture content of samples taken at the end of drying, Moisture Content Variation was ± 0.29 %.

PURPOSE AND SCOPE OF TEST

The Corn Postharvest Processing Plant was tested to determine the operating performance of its major processing facilities, i.e., dehusked corn dryer, corn sheller and grain dryer.

For the dryers the test was conducted to verify and assess the following:

- 1.0 Drying System Performance
 - 1.1 Drying Capacity
 - 1.2 Moisture Reduction per Hour
 - 1.3 Heat Utilization
 - 1.4 Drying Efficiency
 - 1.5 Drying System Efficiency
 - 1.6 Electric Power Consumption
- 2.0 Heating System
 - 2.1 Heating System Efficiency
 - 2.2 Fuel Consumption
- 3.0 Quality of Dried Grains
 - 3.1 Cracked Grain
 - 3.2 Broken Grain
 - 3.3 Moisture Content Variation

For the corn sheller, the test was conducted to assess and verify the following:

- 1.0 Shelling capacity
- 2.0 Quality of work
 - 2.1 Purity of shelled grain
 - 2.2 Percent cracked grain
 - 2.3 Percent broken grain

- 3.0 Shelling recovery
- 4.0 Shelling efficiency
- 5.0 Shelling losses
 - a. Blower loss
 - b. Separation loss
 - c. Unshelled loss
 - d. Scattering loss

PLACE AND DATE OF TEST

The Corn Postharvest Processing Plant was tested as installed at Brgy. Camansi, Alamada, North Cotabato on January 9-11, 2013

METHODS OF TEST

The Dryers and Corn Sheller were tested based on the Philippine Agricultural Engineering Standards “PAES 202:2000 Agricultural Machinery – Heated-Air Mechanical Grain Dryer - Methods of Test” and “PAES 209:2000 Agricultural Machinery – Power-operated Corn Sheller – Methods of Test”, respectively.

OPERATOR’S MANUAL

The operator’s manual submitted included the following: Safety Precautions, Pre-operation Procedure, General Procedure for Loading the Dehusked Corn Dryer, Ear Corn Drying Procedures, Dehusked Corn Dryer Unloading and Shelling Procedure, Grain Drying Procedure using the Re-circulating Dryer, Transfer Grains to Holding Bin Procedure, Trouble Shooting, and Specifications.

Description of the Postharvest Processing Plant

Figure 1 shows the 100-ton input capacity/day ear corn postharvest processing plant. The main components of the processing plant were the following: grain handling system, dehusked corn dryers, corn sheller, grain dryers and the control panel. Newly harvested ear corns were initially dried in the dehusked corn dryer up to a moisture content of around 20%_{w.b.} then shelled and finally dried in grain dryers to a final moisture content of 14 %_{w.b.}. The process flow diagram of the processing plant is shown in **(Fig. 2)**.

Receiving Intake Hopper (Fig. 3)

The ear corn receiving intake hopper was constructed at floor level. It had a capacity of one ton and the bottom was inclined at about 45° for self-emptying. It was made of steel bended plate for self stiffening purposes. The receiving intake hopper was designed to accept an entrance belt conveyor at the bottom.

Entrance Belt Conveyor (Fig. 4)

The entrance belt conveyor was installed at inclined position to convey ear corn from receiving intake hopper to the ear corn dryer. Belts were 450 mm wide and made of ply nylon conveyor belting. It had bended steel plate paddle scrapper spaced at 500 mm on centers for more efficient conveying. Its casing was made out of folded steel plates with heavy duty frame support. Belts were mounted on Engineering plastic idler and return roller.

Loading Belt Conveyor with Diverters (Fig. 5)

The loading belt conveyor with diverters was designed for horizontal conveying of ear corn. Casing was made of folded steel plate with heavy duty frame support. Belt was 450 mm wide and made of ply nylon conveyor belting. Mounted on two 300 mm diameter rollers and equipped with screw-type adjustment of belt tension

at one end, the belt was designed to collect ear corn from the entrance belt conveyor and feed them to the ear corn dryer bins.

Dehusked Corn Dryer

The dehusked corn dryer consisted of the following major components: ear corn drying bins, blower, heat exchanger, and biomass boiler (cob furnace).

Dehusked Corn Drying Bins (Fig. 6)

The dehusked corn dryer consisted of four ear corn drying bins installed in series and constructed out of steel folded plates with a capacity of 50 tons per bin each with dimensions of 7 m x 6 m x 5.5 m. Flooring was made of perforated sheet with steel angular joints and frames designed to hold the total load. Ducts from blower were mounted at the bottom and at the top of the dehusked corn bin for reversible air flow to attain even drying. It had an outlet chute for used air.

Blower (Fig. 7a)

The blower was a radial centrifugal type which was made of impellers, casing, bearing and frame. It was designed to operate at low speed. Airflow was ducted to the ear corn dryer bin.

Heat Exchanger (Fig. 7b)

It was used to transfer heat coming from biomass boiler to the volume of air sucked by the blower. It was made of copper tubes and aluminum cooling fins.

Biomass Boiler (Cob Furnace) (Fig. 8)

Two biomass boilers (cob furnace) were installed for the dehusked corn dryers (one boiler for two dehusked corn drying bins). The biomass boiler was made of boiler tubes and mild steel plates. It was insulated with asbestos cement and

provided with chimney. This biomass boiler was designed to use cobs, woods, or other materials as fuel. Hot water produced by this boiler was pumped to the heat exchanger mounted at the suction side of the boiler. Air temperature was increased through the heat exchanger. The biomass boiler had the following major components: force draft blower (**Fig. 9**), induce draft blower (**Fig. 10**), water pumpset (**Fig. 11**), and valves, piping and accessories.

Dehusked Corn Unloading Belt Conveyor (Fig. 12)

This conveyor was designed for efficient horizontal conveying of ear corn. Casing was made of folded steel plate with heavy duty frame support. Belt was 450 mm wide made of ply nylon conveyor belting. Mounted on two 300 mm diameter end rollers and equipped with two screw-type adjustment of belt tension at one end, the belt was designed to collect ear corn from the drying bin and feed them to the sheller feeder.

Corn Sheller Feeder Belt Conveyor (Fig. 13)

This conveyor was installed at inclined position to convey ear corn from the unloading belt conveyor to the corn sheller. Belts were 450 mm wide made of ply nylon conveyor belting. It had bended steel plate paddle scraper spaced at 500 mm at the center. Its casing was made of folded steel plate with heavy duty frame support. Belts were mounted on bolted bearing idler and return roller.

Corn Sheller (Fig. 14)

The corn sheller was located at the end of the ear corn unloading belt conveyor with an automatic self-feeding conveyor. The sheller operated at low speed. It consisted of a shelling cylinder, an oscillating sieve (**Fig. 15**) which was used to separate the corn kernel from the cob, and aspiration blower (**Fig. 16**) which sucked

light impurities and blew corn cob to the corn cob storage (**Fig 17**) which hold corn cobs that served as fuel for the boiler.

Elevators (**Fig. 18**)

The plant employed two belt-bucket centrifugal discharge type elevators used to convey grains from the corn sheller to the grain dryers and from the grain dryers to the bagging bin.

Grain Dryers

Two batch type recirculating grain dryers were installed with a holding capacity of 16 tons per drying bin. The grain dryers consisted of grain dryer bins with blowers, heat exchangers and a biomass boiler.

Grain Dryer Bin (**Fig. 19**)

The grain dryer bin was cylindrical type and made of perforated sheet Ga. #18 with 4 mm perforations. It had a holding capacity of 16 tons per bin. Heat source was similar to the ear corn drying system which used heat exchanger, biomass boiler and its accessories.

Blower (**Fig. 20a**)

This blower was a radial centrifugal type which was made of heavy duty impeller, casing, bearing and frame. It was designed to operate at low speed. Airflow was ducted to the grain dryer bin.

Heat Exchanger (**Fig. 20b**)

It was used to transfer heat coming from biomass boiler to the volume of air sucked by the blower. It was made of copper tubes and aluminum cooling fins.

Biomass Boiler (Cob Furnace) (Fig. 21)

A biomass boiler (cob furnace) was installed for the two grain dryers. The biomass boiler was made of boiler tubes and mild steel plates. It was insulated with asbestos cement and provided with chimney. This biomass boiler was designed to use cobs, woods, or other materials as fuel. Hot water produced by this boiler was pumped to the heat exchanger mounted at the suction side of the boiler. Air temperature was increased through the heat exchanger. The biomass boiler had the following major components: force draft blower, induce draft blower, water pumpset, and valves, piping and accessories.

Grain Holding Bin (Fig. 22)

This was used as temporary storage of grains before bagging. It was constructed of mild steel plate and its bottom was inclined at an angle of about 35 degrees to assure complete emptying. It was equipped with manual shutters at the bottom.

Control Panel (Fig. 23)

The control panel housed all related motor control components such as magnetic starters, alarm system, circuit breakers and other necessary controls.



Figure 1. Corn Postharvest Processing Plant (Alamada Site)

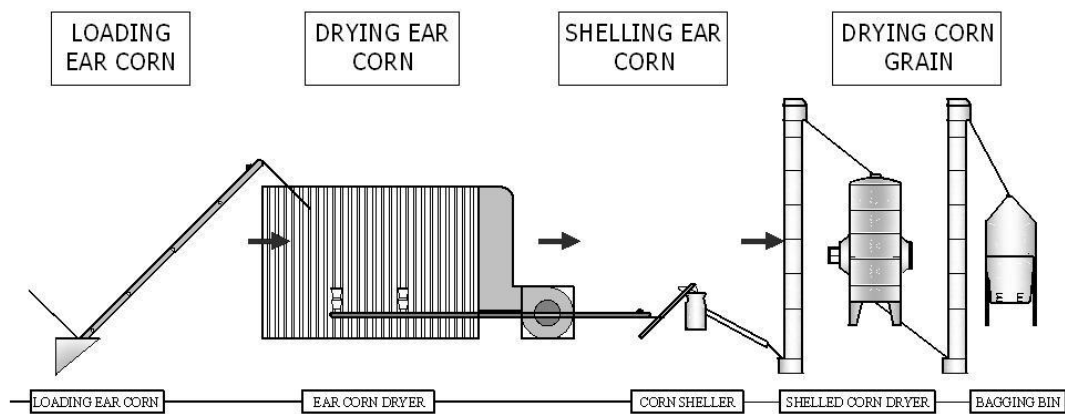


Figure 2. Process Flow of the Processing Plant



Figure 3. Receiving Intake Hopper



Figure 4. Entrance Belt Conveyor

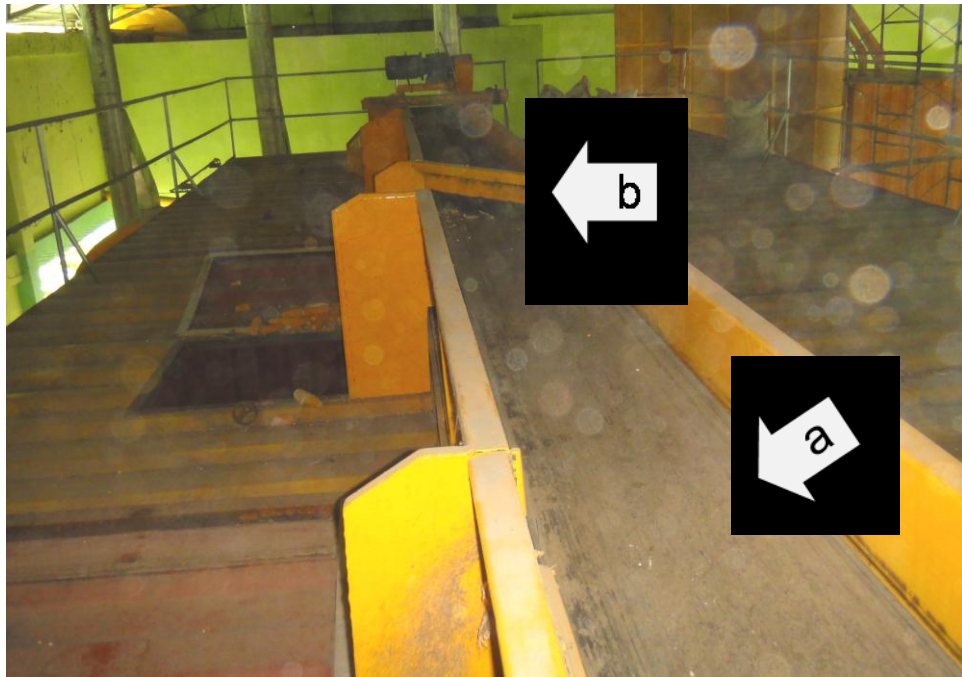


Figure 5. Loading Belt Conveyor (a) with Diverter (b)



Figure 6. Dehusked Corn Drying Bins

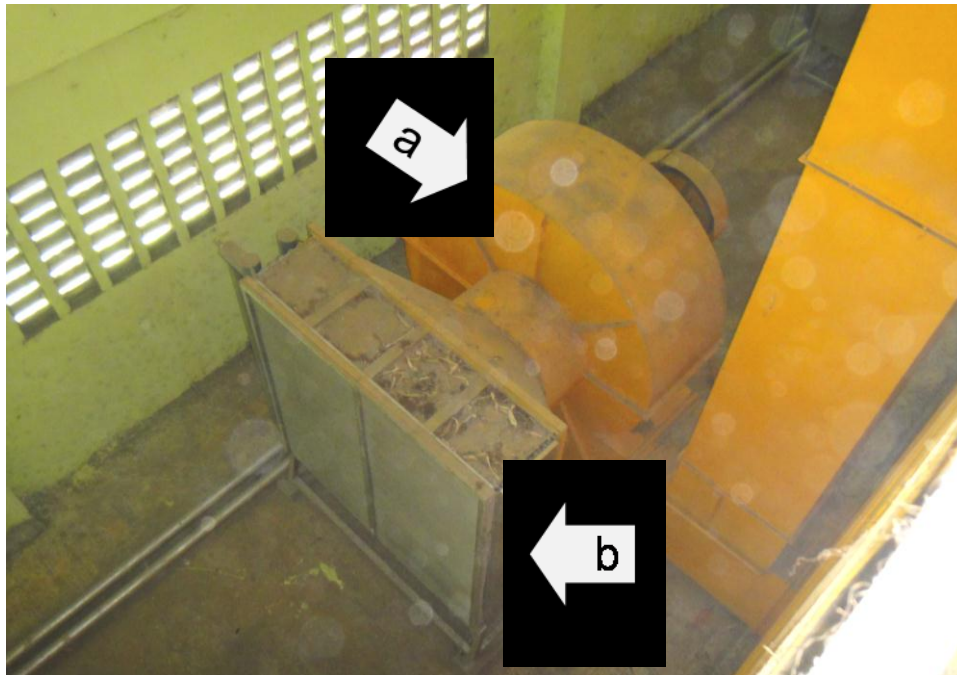


Figure 7. a) Blower
b) Heat Exchanger



Figure 8. Dehusked Corn Dryer's Biomass Boilers



Figure 9. Forced Draft Blower



Figure 10. Induce Draft Blower

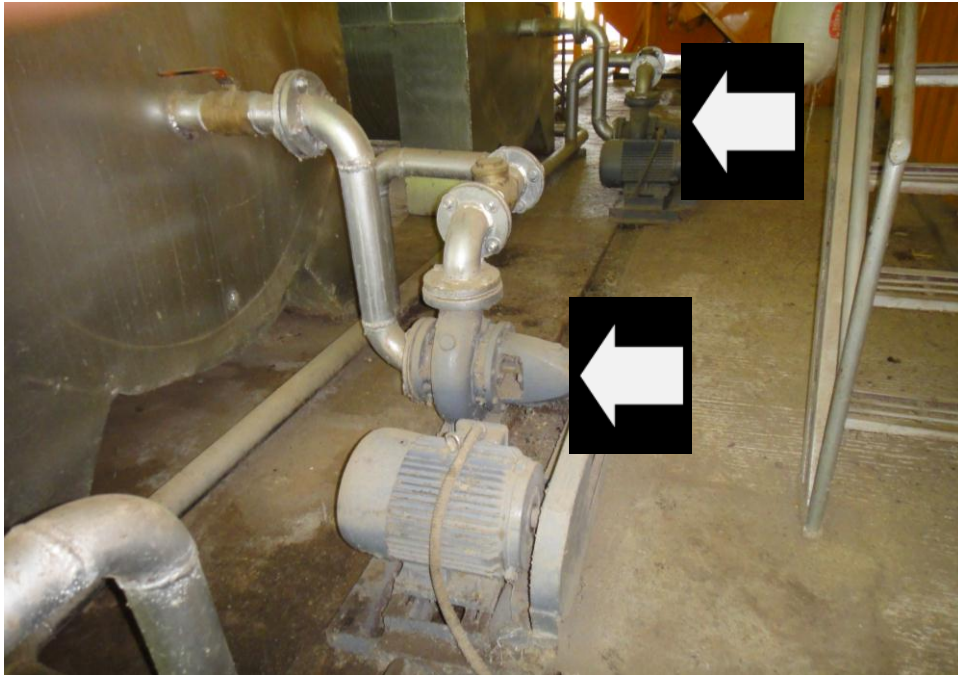


Figure 11. Circulation Pumps

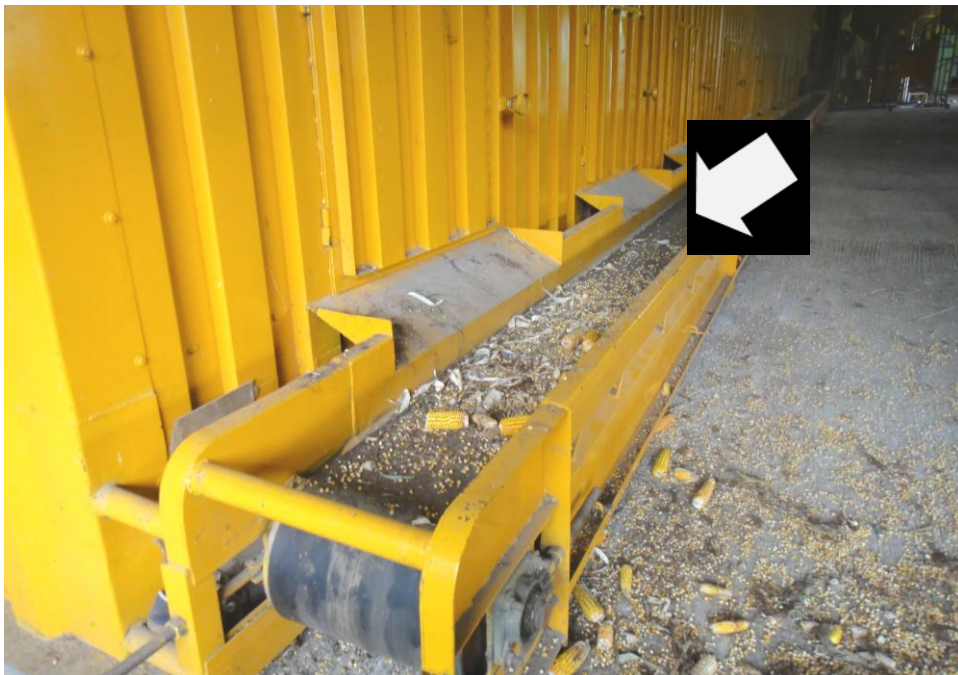


Figure 12. Ear Corn Unloading Belt Conveyor



Figure 13. Corn Sheller Feeder Belt Conveyor



Figure 14. Corn Sheller



Figure 15. Oscillating Sieve

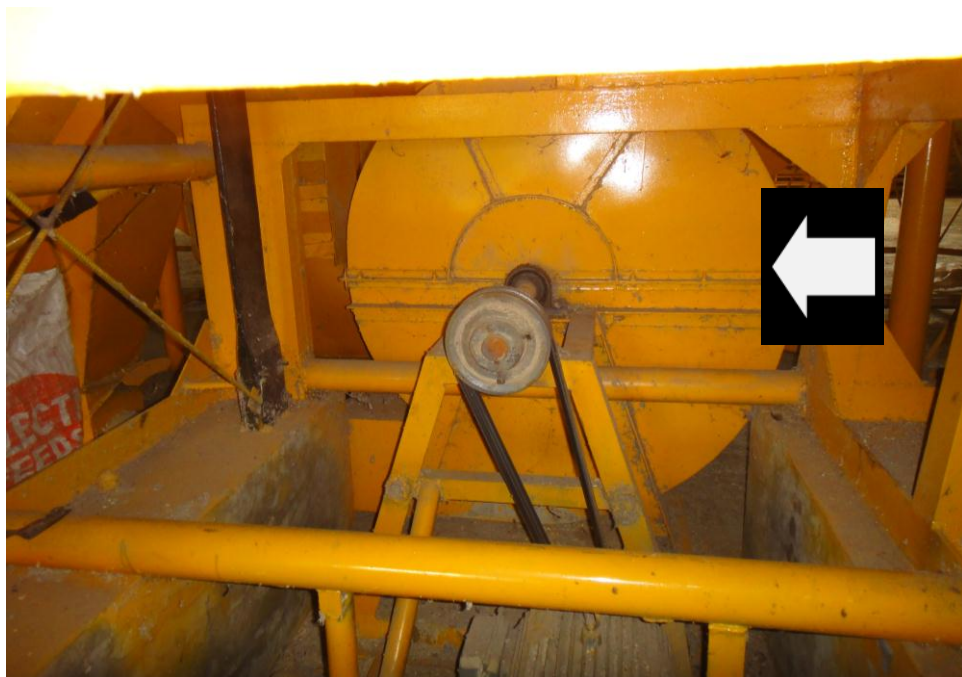


Figure 16. Aspiration Blower

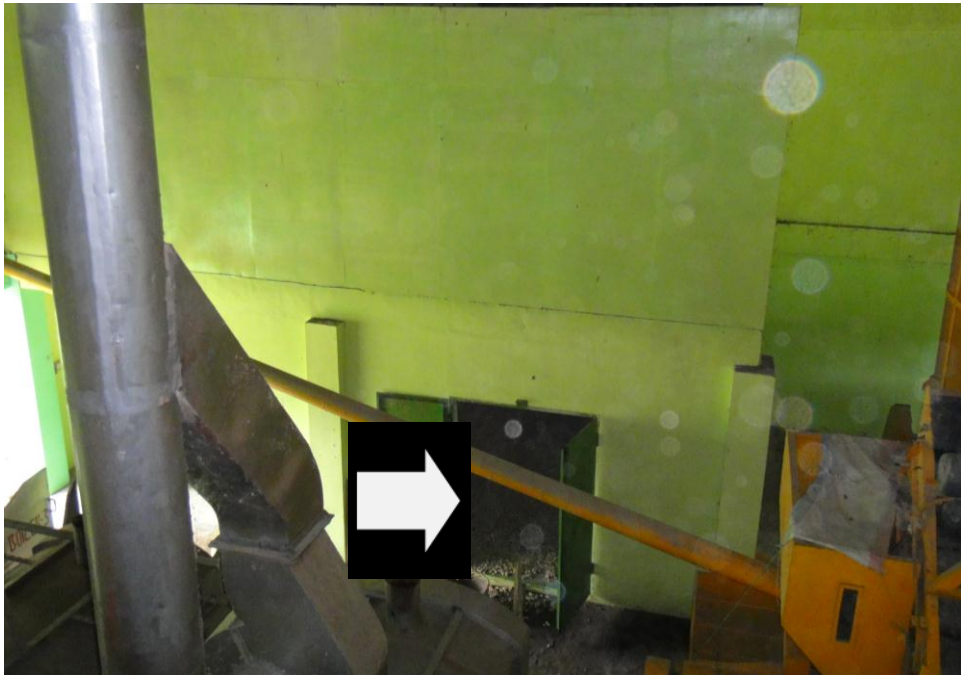


Figure 17. Cob Storage

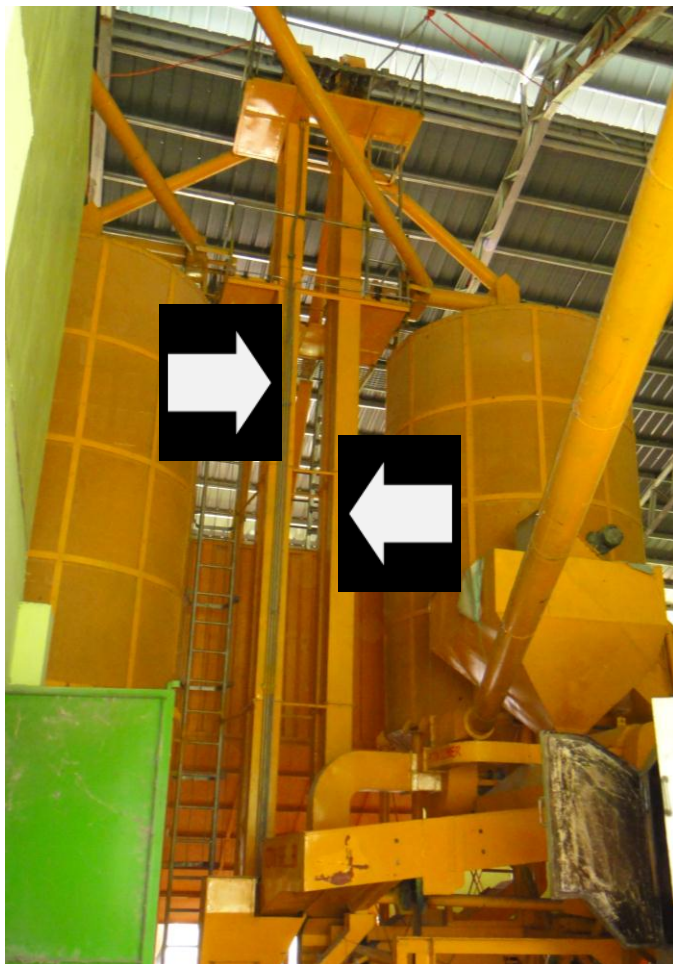


Figure 18. Elevators



Figure 19. Grain Dryer Bins



Figure 20. a) Blower
b) Heat Exchanger



Figure 21. Grain Dryer's Biomass Boiler

Figure 22. Grain Holding Bin





Figure 23. Control Panel

RESULTS AND DISCUSSION

The Plant was a postharvest processing plant for corn with an input capacity of 100 tons of ear corn per day. It was designed to dry newly harvested ear corn to moisture content of around 20%_{w.b} using a dehusked corn dryer (module of two 50-ton ear corn bins and a biomass furnace). The dried ear corn was shelled using corn sheller and then dried to a final moisture content of 14%_{w.b} in grain dryer (module of two 16-ton grain dryer bins and a biomass furnace). The plant was tested for performance using 99836 kg of ear corn. Corn cobs were used as fuel for the heating system.

DEHUSKED CORN DRYER (Model: CBD-2005)

Results of performance tests are shown in **Table 1**. Refer to **Annex 1** for formula used in the computation of results.

Drying Capacity

The drying capacity of the dryer was the amount of fresh ear corn dried to the desired moisture content per unit of time. The dehusked corn dryer had a Drying Capacity of 2812 kg/h from an initial moisture content of 27.7%_{w.B}. to a moisture content of 19.1%_{w.B}.

Moisture Content Reduction

The average Moisture Content Reduction during the drying process was 0.24% per hour.

Moisture Content Variation

Based on the moisture content of samples taken from the unloading horizontal conveyor of the ear corn dryer, Moisture Content Variation at the termination of drying was $\pm 0.46\%$.

Heating System Efficiency

Heating system efficiency was the ratio of the amount of heat supplied for drying to the total amount of heat available from the fuel, expressed in percent. The dryer had Heating System Efficiency of 78.5%.

Furnace's Fuel Consumption

The fuel consumption of the furnace was determined by getting the total amount of fuel consumed during the test divided by the total time of operation. The Fuel Consumption of the furnace was 197.8 kg/h of corn cobs.

Power Consumption

The Total Power Consumption of the dryer was 19.0 kW with Load Current of 39.3 A at a Line Voltage of 472 V.

Observations

The dryer operated smoothly and no breakdown or failure of components occurred during the test. The following were also observed:

1. The top portion of the drying bins was provided with railing (**Fig. 24**).
2. The power transmission of the blower was also provided with cover (**Fig. 25**).



Figure 24. Railing installed on top of the drying bins

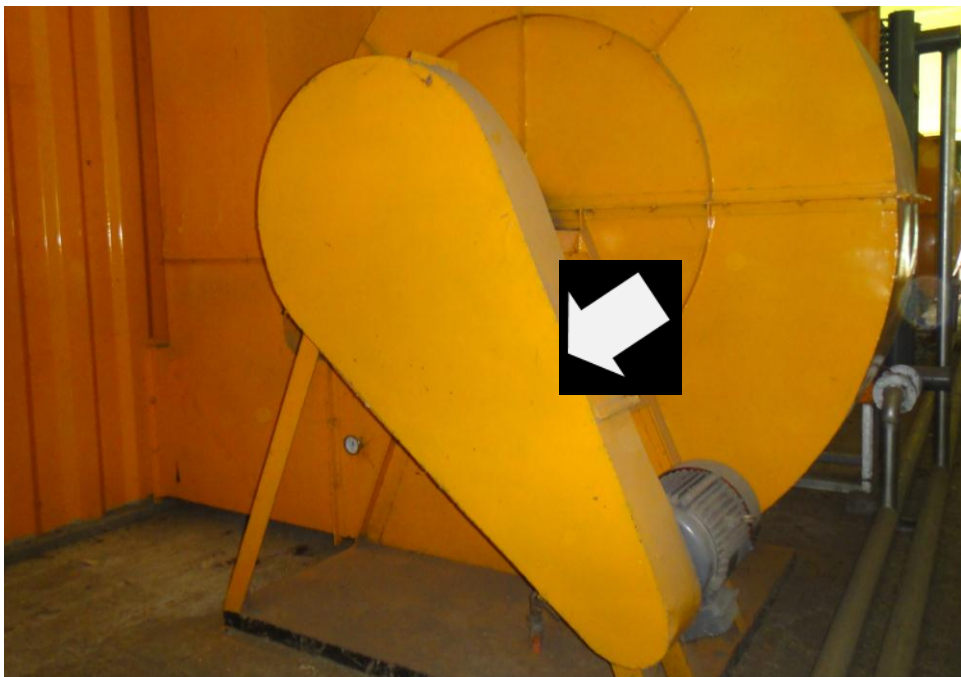


Figure 25. Belt cover

Table 1. Results of Performance Test of the Dehusked Corn Dryer (two bins)

ITEMS	DATA
1. Dryer's Total Ear Corn Holding Capacity, two bins (kg)	100000
2. Initial Weight of Test Material (kg)	99836
4. Initial Moisture Content (%)	27.7
5. Final Moisture Content (%)	19.1
6. Actual Drying Time (h)	35.5
7. Drying Capacity (kg/h)	2812
9. Moisture Reduction Rate (%/h)	0.24
10. Average Drying Air Temperature (°C)	66.8
11. Ambient Air Temperature (°C)	
a) Dry Bulb	29.7
b) Wet Bulb	26.7
12. Ambient Air Relative Humidity (%)	79.2
13. Dryer Exhaust Air Temperature (°C)	
a) Dry Bulb	36.7
b) Wet Bulb	34.0
14. Exhaust Air Relative Humidity (%)	83.2
15. Speed of Components (rpm)	
a) Fan's Motor	1785
b) Fan	303
16. Static Pressure between heat exchanger and blower (mmH ₂ O)	34
17. Fan Air Velocity (m/s)	3.35
18. Air Flow Rate (m ³ /min)	625.1
19. Plenum Static Pressure (mmH ₂ O)	6
20. Furnace Consumption Rate, corn cob (kg/h),	197.8
21. Heating System Efficiency (%)	78.5
22. Noise Level at Corn Cob Feeder [db(A)]	No Data*
23. Average Power Consumption during drying process	
a) Electrical Input Power (kW)	19.0
b) Line Voltage (V)	472
c) Load Current (A)	39.3
24. Moisture Content Variation at the Termination of Drying (%)	±0.46

* Instrument out of order

CORN SHELLER (Model: CS-2005)

Results of performance tests are shown in **Tables 2 to 5**. Refer to **Annex 2** for formulas used in the computation of results.

Capacity

Test results showed that the machine had a Shelling Capacity of 16113 kg/h using mixed varieties of corn at 19.1% MC_{wb} and Kernel-Ear Corn Ratio of 0.81.

For purposes of standardization, the Corrected Shelling Capacity, based on 20% MC_{wb} and 100% Purity, was 16225 kg/h.

Performance

The performance of the sheller was evaluated in terms of its shelling recovery, shelling efficiency, shelling losses, purity and grain quality.

Shelling efficiency and recovery. The Shelling Efficiency of the machine was 99.91% and the Shelling Recovery was 99.73%

Shelling loss. Total Shelling Loss was 0.27% consisting of 0.16% Scattering Loss, no Blower Loss, 0.02% Separation loss and 0.09% Unshelled loss.

Purity. The Purity of collected shelled grain was 99.70%.

Grain quality. Quality of shelled grain was determined in terms of percent cracked and broken grains. Results showed that the shelled output had a Net Percentage Cracked Grains of 4.67% and Net Percent Broken Grains of 3.45%. (Cracked grains refer to those grains whose endosperms were partially fractured while broken grains refer to those grains that were split into 2 or more pieces.)

Power Consumption

The Total Power Consumption of the sheller was 46.2 kW with Load Current of 83.5 A at a Line Voltage of 462 V.

Observations:

1. The sheller was provided with a stair and a platform where one of the operators stood during shelling operation (**Fig. 26**).
2. The space between the sheller's hopper and the top end of the inclined conveyor was provided with an outlet chute (**Fig. 27**) to collect corn ears that might fall on the ground during shelling operation.



Figure 26. Stair and platform

Figure 27. Outlet chute for corn ear that might fall to the ground

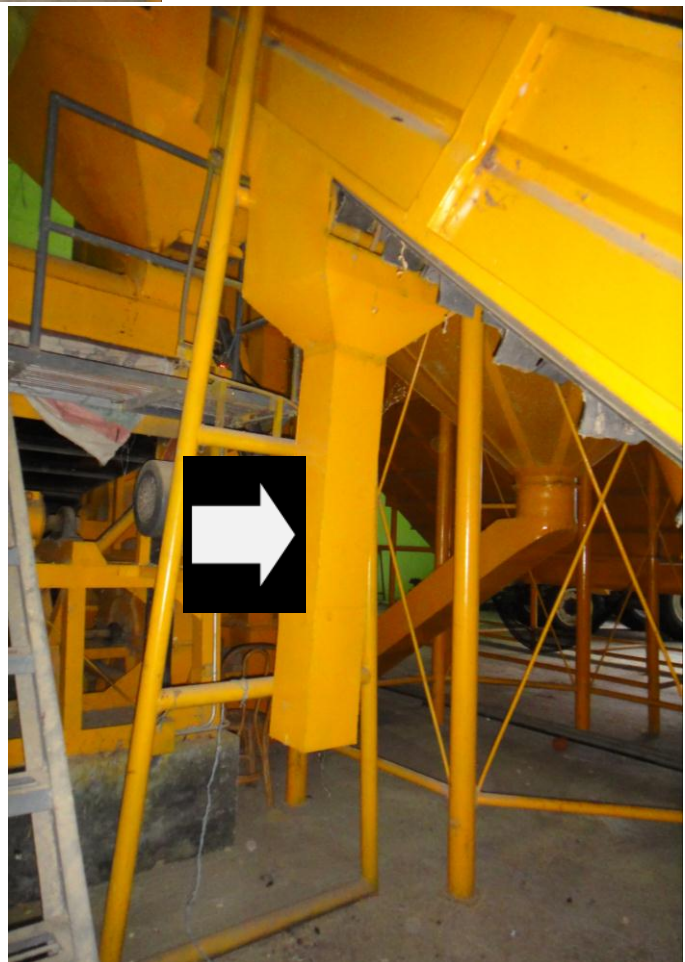


Table 2. Condition of crop used during the test

ITEMS	Data
1. Corn variety	Mixed
2. Days after harvest	5
3. Kernel-Ear Corn Ratio	0.81
4. Size of corn ear, L x D (mm)	166 x 46
5. Grain moisture content (%), w.b.	19.1

Table 3. Speed of sheller components during the test

ITEMS	Data
1. Sheller Main Motor (rpm)	
Without load	1798
With load	1747
2. Shelling Cylinder (rpm)	
Without load	873
With load	836
3. Shelling Cylinder Peripheral Speed (m/min)	
Without load	1397
With load	1338
4. Oscillating Screen (rpm)	
Without load	499
With load	470
5. Cob Thrower Fan (rpm)	
Without load	872
With load	868
6. Aspirating Fan (rpm)	
Without load	1079
With load	1060

Table 4. Sheller performance during the test

ITEMS	Data
1. Sheller capacity (kg/h)	
Actual	16113
Corrected*	16225
2. Purity of shelled grain (%)	99.70
3. Shelling recovery (%)	99.73
4. Shelling efficiency (%)	99.91
5. Blower loss (%)	Nil
6. Separation loss (%)	0.02
7. Unshelled loss (%)	0.09
8. Scattering loss (%)	0.16
9. Total shelling loss (%)	0.27
10. Noise level [db(A)], ear level of the operator who control the feed gate	
Without load	No Data
With load	No Data
11 Average Power Consumption During Shelling Operation	
a) Electrical Input Power (kW)	46.2
b) Line Voltage (V)	462
c) Load Current (A)	83.5

*Adjusted to 20% grain moisture content and 100% purity of shelled grain

Table 5. Laboratory grain analysis of samples gathered during testing

ITEMS	Data*
1. Before Shelling (%)	
Cracked	0
Broken	0
2. After Shelling (%)	
Cracked	4.67
Broken	3.45
3. Net Percent	
Cracked	4.67
Broken	3.45

GRAIN DRYER (Model: GD-2005)

Results of performance tests are shown in **Tables 6 to 8**. Refer to **Annex 1** for formulas used in the computation of results.

Drying Capacity

The drying capacity of the grain dryer is the amount of kernel dried to the desired moisture content per unit of time. The grain dryer had a Drying Capacity of 9069 kg/h from an initial moisture content of 19.1%_{W.B.} to a final moisture content of 13.2%_{W.B.}.

Moisture Content Reduction

The average Moisture Content Reduction during the drying process was 1.68% per hour.

Moisture Content Variation

Based on the moisture content of samples taken from the outlet chute of the dryer, Moisture Content Variation at the termination of drying was $\pm 0.29\%$.

Heating System Efficiency

Heating system efficiency is the ratio of the amount of heat supplied for drying to the total amount of heat available from the fuel, expressed in percent. The dryer had Heating System Efficiency of 86.8%.

Furnace's Fuel Consumption

The fuel consumption of the corn cob furnace was determined by getting the total amount of corn cob consumed during the test divided by the total time of operation. The Fuel Consumption of the furnace was 187.1 kg/h of corn cob.

Heat Utilization

Heat utilization refers to the total amount of heat utilized to vaporize moisture from the grain. Results showed that the dryer had a Heat Utilization of 5684 kJ/kg of water removed.

Drying Efficiency

Drying efficiency is the ratio of the total heat utilized to vaporize moisture in the grain to the amount of heat added to the drying air, expressed in percent. Results showed that the dryer had a Drying Efficiency of 41.2%

Drying System Efficiency

Drying system efficiency is the ratio of the total heat utilized for drying to the heat available in the fuel, expressed in percent. Results showed that the dryer had a Drying System Efficiency of 35.7%.

Power Consumption

The Total Power Consumption of the dryer was 24.6 kW with Load Current of 52.2 A at a Line Voltage of 461 V.

Observations

The dryer operated smoothly and no breakdown or failure of components occurred during the test. The following were also observed.

1. The elevators were provided with intake hoppers where manual loading of grains can be done (**Fig 28**).
2. A cyclone connected to the aspirating fans of the elevators was installed outside the building to collect impurities during the drying operation (**Fig. 29**).

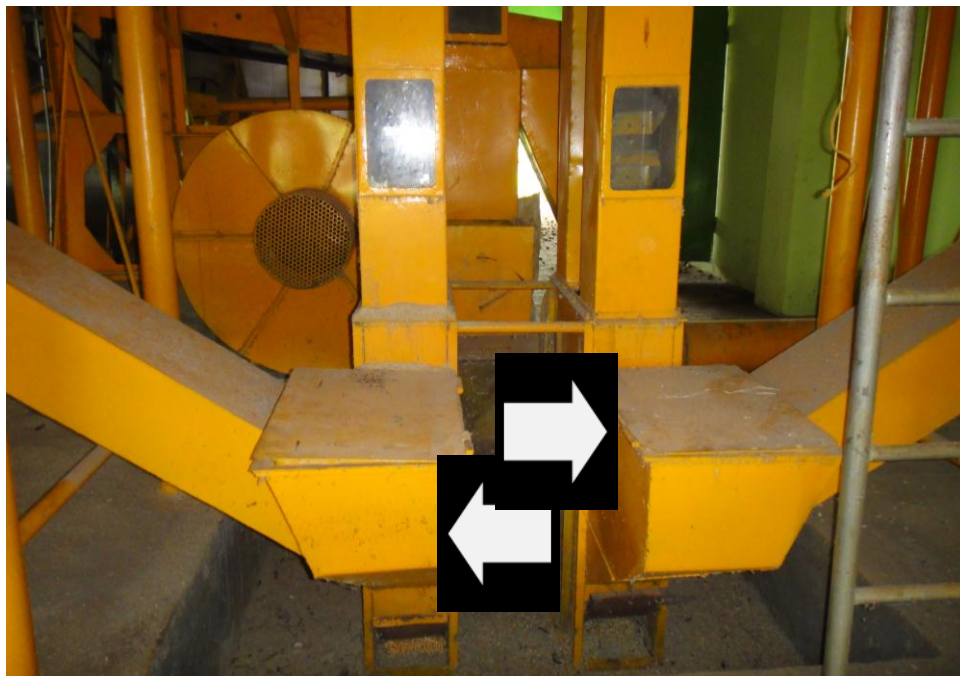


Figure 28. Elevators' intake hoppers



Figure 29. Cyclone for collection of impurities

Table 6. Results of Performance Test of the Grain Dryer (two bins)

ITEMS	DATA
1. Dryer's Total Grain Holding Capacity, two bins (kg)	32000
2. Initial Weight of Test Material (kg)	31743
3. Final Weight of Test Material (kg)	29585
4. Initial Moisture Content (%)	19.1
5. Final Moisture Content (%)	13.2
6. Actual Drying Time (h)	3.5
7. Drying Capacity (kg/h)	9069
8. Rate of Moisture Removal (kg/h)	616.6
9. Moisture Reduction Rate (%/h)	1.68
10. Average Drying Air Temperature (°C)	67.9
11. Ambient Air Temperature (°C)	
a) Dry Bulb	31.5
b) Wet Bulb	26.6
12. Ambient Air Relative Humidity (%)	68.3
13. Dryer Exhaust Air Temperature (°C)	
a) Dry Bulb	37.8
b) Wet Bulb	33.8
14. Exhaust Air Relative Humidity (%)	76.2
15. Grain Temperature at the Drying Section (°C)	41.8
16. Grain Temperature at the Cooling Section (°C)	38.1
17. Speed of Components (rpm)	
a) Fan's Motor	1785
b) Fan	301
18. Static Pressure between Heat Exchanger & Blower (mmH ₂ O)	23
19. Fan Air Velocity (m/s)	3.33
20. Air Flow Rate (m ³ /min)	668.6
21. Plenum Static Pressure (mmH ₂ O)	12
22. Furnace Fuel Consumption Rate, corn cob (kg/h)	187.1
23. Heating System Efficiency (%)	86.8
24. Heat Utilization (kJ/kg moisture removed)	5684
25. Drying Efficiency (%)	41.2
26. Drying System Efficiency (%)	35.7
27. Noise Level at Corn Cob Feeder [db(A)]	No Data

Table 6. Results of Performance Test of the Grain Dryer (two bins), cont'n.

28. Average Power Consumption during drying process	
a) Electrical Input Power (kW)	24.6
b) Line Voltage (V)	461
c) Load Current (A)	52.2
29. Moisture Content Variation at the Termination of Drying (%)	±0.29
30. Total Loading Time (h)	1.97

Table 7. General information of air-dried shelled corn samples brought to AMTEC laboratory for analysis

ITEMS	DATA
Variety	Mixed
Source	Alamada, North Cotabato
Purity (%)	98.92
Bulk Density (kg/m ³)	748.1
Moisture Content, wet basis (%)	13.5
Cracked Grains (%)	3.60
Broken Grains (%)	1.08

Table 8. General information of shelled corn samples dried using the Grain Dryer brought to AMTEC laboratory for analysis

ITEMS	DATA
Variety	Mixed
Source	Alamada, North Cotabato
Purity (%)	99.92
Bulk Density (kg/m ³)	754.6
Moisture Content, wet basis (%)	13.2
Cracked Grains (%)	6.50
Broken Grains (%)	1.50

ANNEX 1. FORMULA USED IN THE COMPUTATION OF DRYING PARAMETERS

$$1. \quad \text{Drying Capacity (kg/h)} = \frac{W_1}{T_A}$$

where,

W_1 = Initial weight of test material (kg)

T_A = Actual drying time (h)

$$2. \quad \text{Initial Weight of Test Materials, (kg)}$$

$$W_1 = \frac{W_2 (100 - MC_2)}{(100 - MC_1)}$$

Where,

W_2 = Final weight of test materials (kg)

MC_1 = Initial moisture content (%)

MC_2 = Final moisture content (%)

$$3. \quad \text{Moisture Reduction Per Hour (kg/h)}$$

$$\text{Moisture reduction per hour} = \frac{W_1 - W_2}{T_A}$$

$$4. \quad \text{Heating System Efficiency, } E_{HS} \text{ (%)}$$

$$E_{HS} = \frac{Q_s}{Q_f} \times 100$$

Where,

E_{HS} = Heating system efficiency (%)

Q_s = Heat supplied to the dryer (kJ/h)

Q_f = Heat available in the fuel (kJ/h)

To solve for Q_s :

$$Q_s = \frac{\Delta h \times Q}{v} \times 60$$

Where,

Δh = Change in enthalpy (kJ/kg d.a.)

Q = Air flow rate (m³/min)

v = Specific volume of fresh air (m³/kg d.a.)

To solve for Q_f :

$$Q_f = F_c \times H_{vf}$$

Where,

F_c = Fuel consumption (kg/h)

H_{vf} = Heating value of the fuel used (kJ/kg)

5. Heat utilization, Q_u (kJ/kg)

$$Q_u = \frac{Q_s \times T_A}{W_1 - W_2}$$

6. Drying Efficiency (%)

$$E_D = \frac{Q_R}{Q_S} \times 100$$

Where,

E_D = Drying efficiency (%)

Q_R = Heat required to vaporize moisture inside the grain (kJ/h)

To solve for Q_R

$$Q_R = Q_v - (W_1 - W_2)/T_A$$

Where,

Q_v = Heat of vaporization of water

7. Drying System Efficiency, E_{DS} (%)

$$E_{DS} = Q_R/Q_f \times 100$$

Annex 2. FORMULA USED IN THE COMPUTATION OF SHELLING PARAMETERS

1. Kernel-ear corn ratio = $\frac{W_K}{W_{EC}}$ where:

W_K = weight of corn kernel (kg)

W_{EC} = weight of ear corn, w/o husk (kg)

2. Fuel consumption

$$F_C = \frac{F_V}{T_A} \quad \text{where:}$$

F_C = Fuel consumption (L/h)

F_V = Total amount of fuel consumed (L)

T_A = Time the engine is in operation (h)

3. Shelling capacity

$$C_A = \frac{W_{SG}}{T_E} \quad \text{where:}$$

C_A = Actual shelling capacity (kg/h)

W_{SG} = Weight of shelled grain (kg)

T_E = Time elapsed

4. Corrected shelling capacity

$$C_C = \frac{100 - MC_O}{100 - MC_M} \times P \times C_A \quad \text{where:}$$

C_C = Corrected shelling capacity (kg/h)

C_A = Actual shelling capacity (kg/h)

MC_O = Observed grain moisture content (%)

MC_M = Kernel moisture content (%), at 20%

P = Kernel purity (%)

5. Purity

$$P = 100\% - \% \text{Impurity}$$

6. Impurity

$$I = \frac{G_{US} - G_{CS}}{G_{US}} \times 100 \quad \text{where:}$$

I = Impurity (%)

G_{US} = Uncleaned grain sample (g)

G_{CS} = Cleaned grain sample (g)

7. Shelling Efficiency

$$N_S = \frac{W_{SG} + L_S + L_B}{W_{SG} + L_S + L_B + L_U} \times 100 \quad \text{where:}$$

N_S = Shelling Efficiency (%)

L_S = Separation loss (kg)

L_B = Blower loss (kg)

L_U = Unshelled loss (kg)

8. Shelling Recovery, percent

$$R_S = \frac{W_{SG}}{W_{SG} + L_S + L_B + L_U} \times 100$$

9. Unshelled loss, percent

$$PL_U = \frac{L_U}{W_{SG} + L_S + L_B + L_U} \times 100$$

10. Separation loss, percent

$$PL_S = \frac{L_S}{W_{SG} + L_S + L_B + L_U} \times 100$$

11. Blower loss, percent

$$PL_B = \frac{L_B}{W_{SG} + L_S + L_B + L_U} \times 100$$

12. Cracked grain

$$PG_C = \frac{G_C}{100 \text{ grainsample}} \times 100 \quad \text{where:}$$

PG_C = Percent cracked grain

G_C = Number of cracked grain

13. Broken grain

$$PG_B = \frac{G_B}{100 \text{ gram sample}} \times 100 \quad \text{where:}$$

PG_B = Percent broken grain

G_B = Weight of broken (g)

14. Net Cracked grains

$$NG_C = IG_C - FG_C \quad \text{where;}$$

NG_C = Net Cracked Grains

IG_C = Number of Cracked grains before shelling

FG_C = Number of Cracked grains after shelling

15. Net Broken grains

$$NG_B = IG_B - FG_B \quad \text{where;}$$

NG_B = Net Broken Grains

IG_B = Weight of Broken Grains before shelling (g)

FG_B = Weight of Broken Grains after shelling (g)

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Checked by:

ROMULO E. EUSEBIO

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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